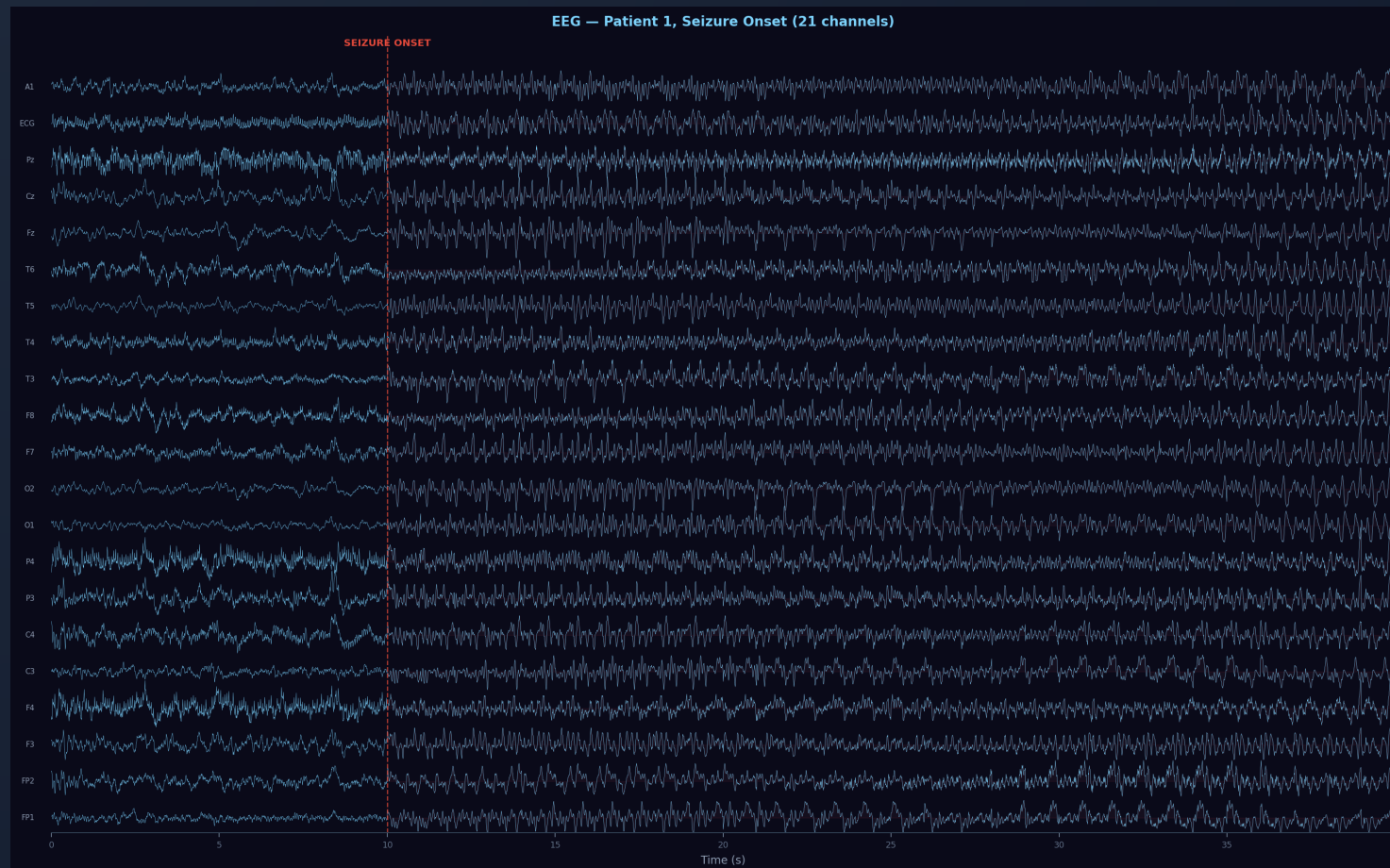


# Deep Learning for EEG Seizure Detection



# Contents

---

1. Introduction & motivation
2. Dataset & analysis
3. Preprocessing & data flow
4. Methodology: 4 pipelines
5. Cross-validation strategies
6. LSTM design: pooling & context
7. Hyperparameters & experiment grid
8. Results & conclusion

# Introduction

---

Epileptic seizures = abnormal brain electrical activity detectable via EEG.

**Goal:** automatically detect seizures in EEG recordings.

**Approach:** compare progressively richer models

# Introduction - Models

---

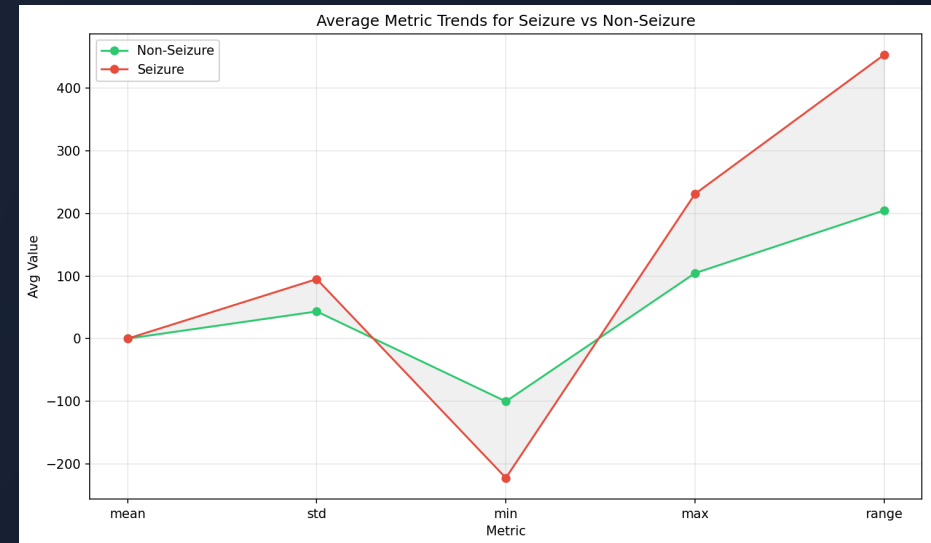
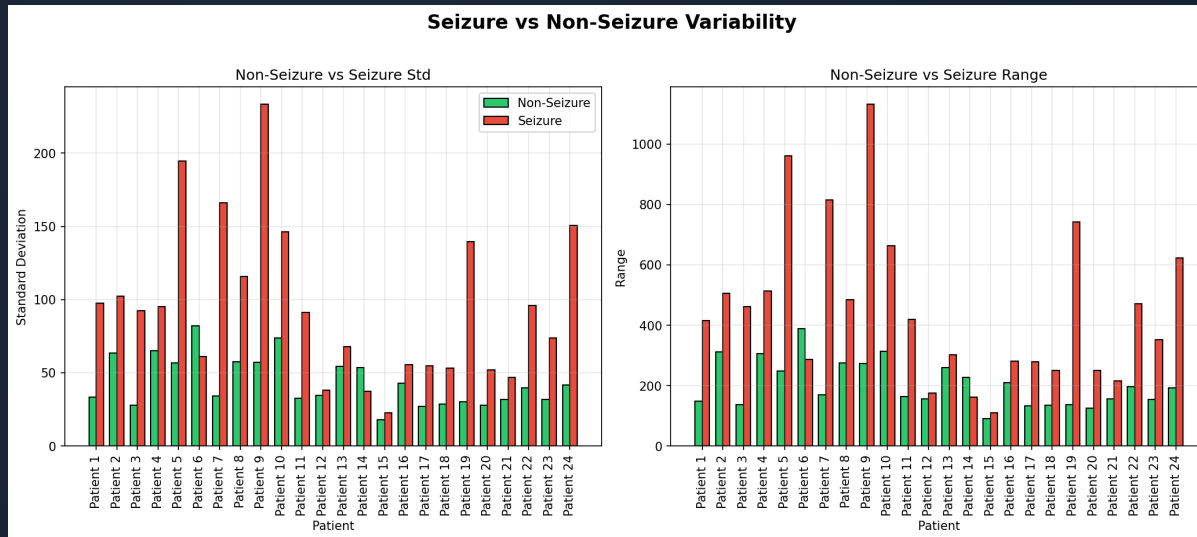
1. Threshold baseline (hand-crafted rules)
2. Random forest (hand-crafted features)
3. CNN (learned features, end-to-end)
4. LSTM (temporal episode modelling)

# Dataset: CHB-MIT

---

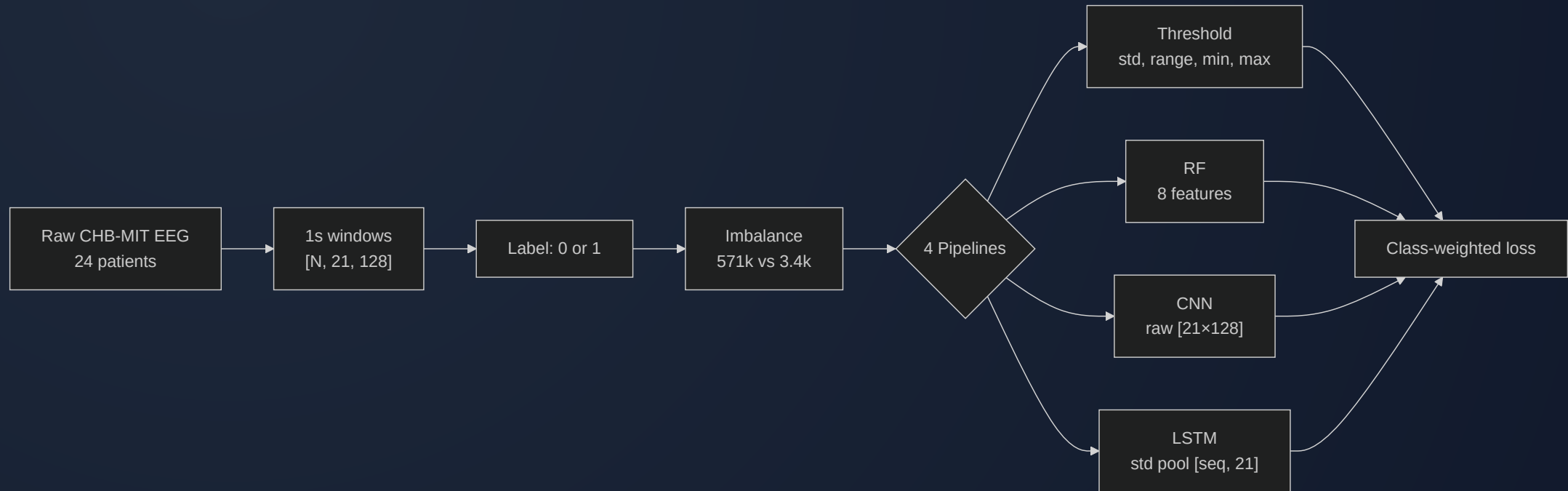
- **Source:** CHB-MIT scalp EEG, 24 pediatric patients
- **Windows:** 1s segments → **571,905** total
- **Seizure:** **86,672** windows (15.2% — imbalanced dataset)
- **Shape:** each window = **21 channels × 128 timesteps**

# Dataset: Feature Trends



- **Std** and **range** separate seizure from non-seizure
- **Mean**  $\approx 0$  for both  $\rightarrow$  risky feature alone

# Preprocessing & Data Flow



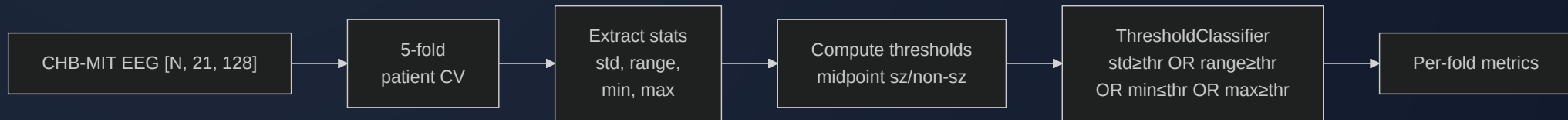
- Reusing existing augmentation from the dataset
- No per-window normalization
- **Class weighting** helps models focus on seizure

# Overview: 4 Pipelines



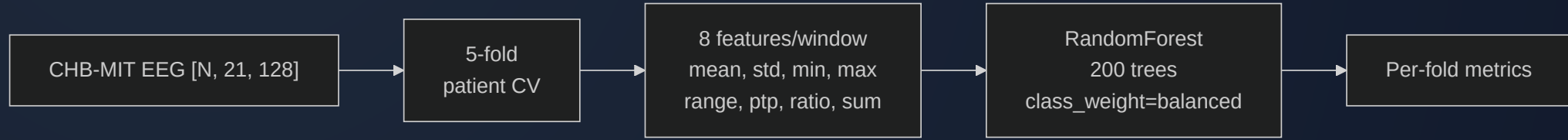


# Pipeline 1: Threshold Classifier



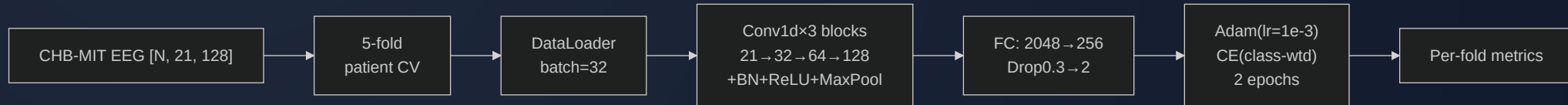
- Computes **std, range, min, max** per window
- Flags seizure if stat exceeds fold-specific threshold
- Thresholds = midpoint of seizure/non-seizure means

# Pipeline 2: Random Forest



- **8 features:** mean, std, min, max, range, ptp, std/range ratio, range+std
- 200 trees
- Patient-level 5-fold CV

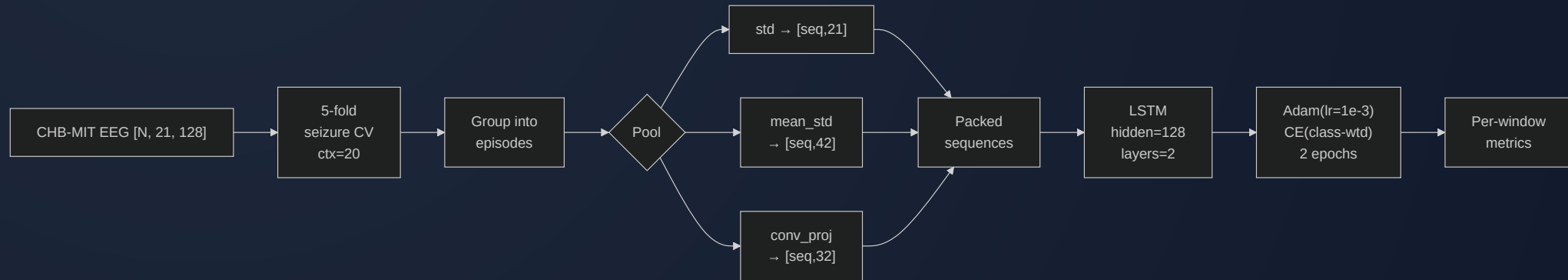
# Pipeline 3: CNN



- **Input:** raw window [21, 128], no normalization
- 3× Conv1d → BN → ReLU → MaxPool
- FC: 2048→256→Dropout(0.3)→2
- Adam (lr=1e-3), class-weighted CE, 2 epochs

# Pipeline 4: LSTM

*Adapted from course-provided EpilepsyLSTM  
(ChakrabartiChannelFusion)*



- **Per-timestep classification**
- **Conv1d projection** option + packed sequences for variable-length episodes
- Per-window **pooling** reduces [21, 128] → feature vector

# Cross-Validation Strategies

## Seizure-level

Whole episode → one fold

$\pm 20$  context windows in val

## Window-level

Window round-robin

$\pm 4$  seizure neighbors  
dropped

# Seizure-Level k-fold: Context

---

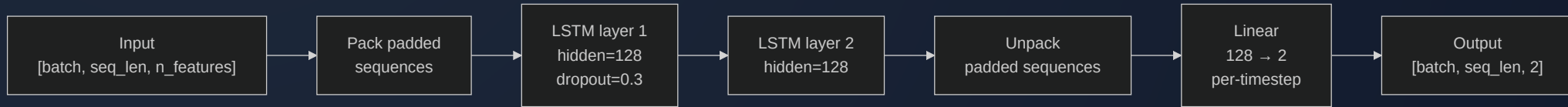
- Without context: val = **100% seizure** → no specificity measurable
- With `context=20`:  $\pm 20$  non-seizure windows around each episode
- Val gets realistic  $\sim 40\text{--}60\%$  non-seizure mix
- Train gets all remaining windows

# LSTM: Pooling Options

EEG oscillates around 0 → **mean** collapses amplitude. **Std** preserves it.

|  | Pool          | Dim  | Speed     | Preserves         |
|--|---------------|------|-----------|-------------------|
|  | std (default) | 21   | Fast      | Channel variance  |
|  | mean          | 21   | Fast      | Near-zero — risky |
|  | mean_std      | 42   | Fast      | Center + spread   |
|  | conv_proj     | 32   | Medium    | Learned features  |
|  | none          | 2688 | Very slow | Full waveform     |

# LSTM: Architecture



- Packed sequences for variable-length episodes
- Class-weighted CE



# LSTM: conv\_proj Mode

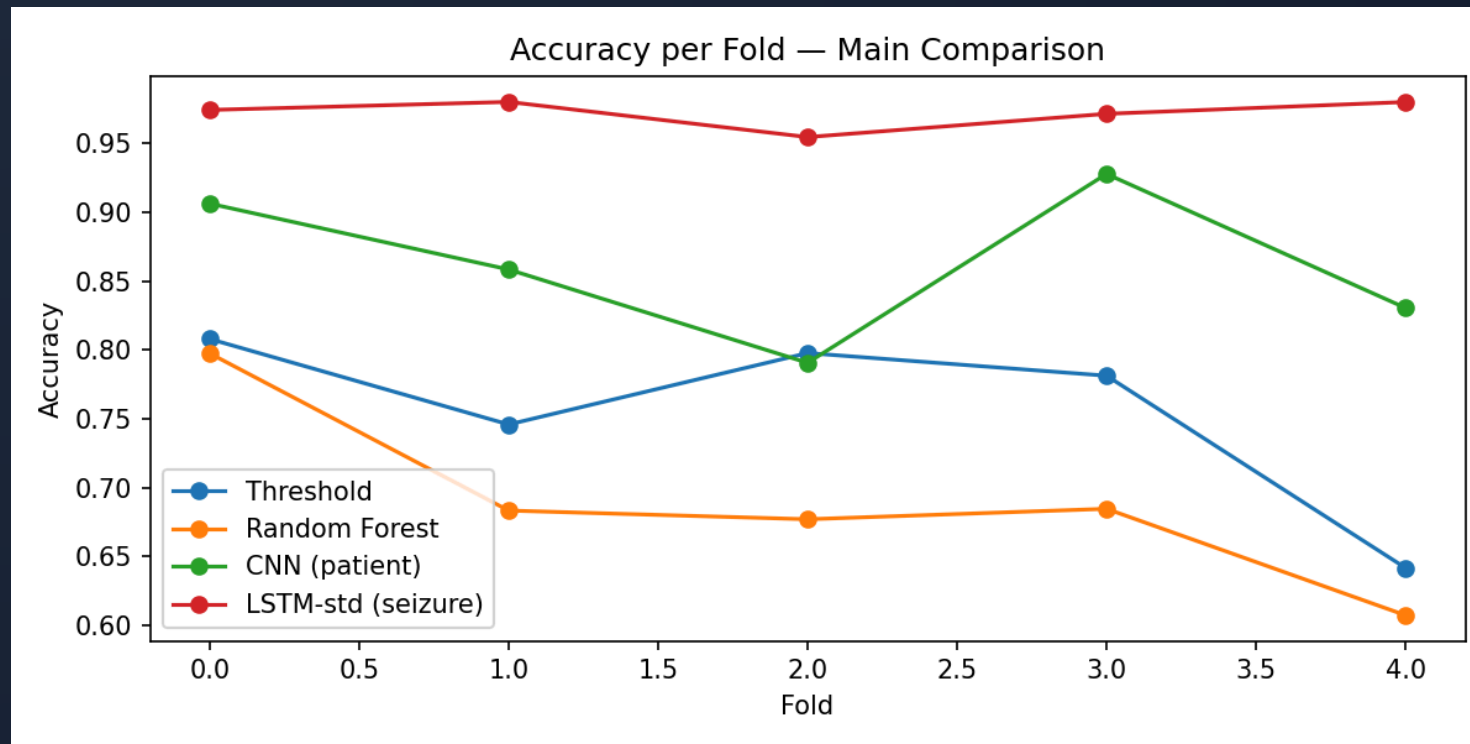


- Learns waveform features end-to-end
- Faster than **none** (2688-d), more expressive than **std** (21-d)

# Hyperparameters

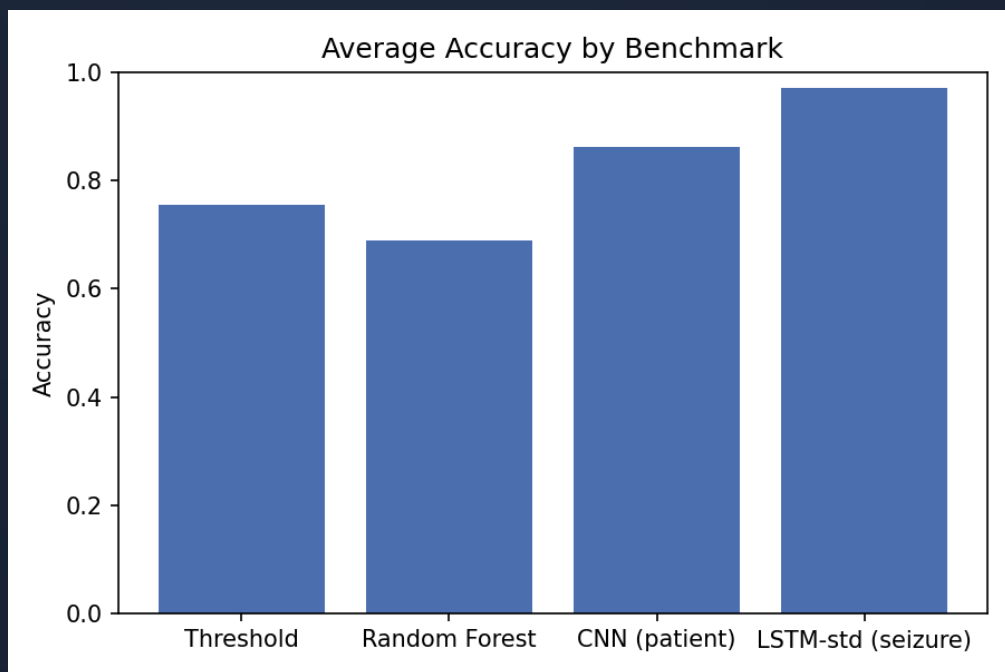
| Parameter    | Threshold | RF       | CNN         | LSTM             |
|--------------|-----------|----------|-------------|------------------|
| Trees/layers | —         | 200      | [32,64,128] | 2 LSTM           |
| Hidden size  | —         | —        | —           | 128              |
| Pooling      | —         | —        | —           | std / conv_proj  |
| Class weight | —         | balanced | balanced    | balanced         |
| LR           | —         | —        | 1e-3        | 1e-3             |
| Batch size   | —         | —        | 32          | 32               |
| Epochs       | —         | —        | 2           | 2                |
| Dropout      | —         | —        | 0.3         | 0.3              |
| CV level     | patient   | patient  | patient     | seizure (ctx=20) |

# Results: Accuracy per Fold (Patient-Level)



High cross-patient variance across all models;  
CNN ranges 79–93%.

# Results: Main Comparison (5-fold)



| Model     | CV      | Acc          | F1           |
|-----------|---------|--------------|--------------|
| Threshold | patient | 75.5%        | —            |
| RF        | patient | 69.0%        | —            |
| CNN       | patient | 86.3%        | 0.554        |
| CNN       | window  | 97.7%        | 0.000        |
| CNN       | seizure | 80.1%        | 0.888        |
| LSTM-std  | seizure | <b>97.2%</b> | <b>0.985</b> |
| LSTM-std  | patient | 99.5%        | 0.983        |

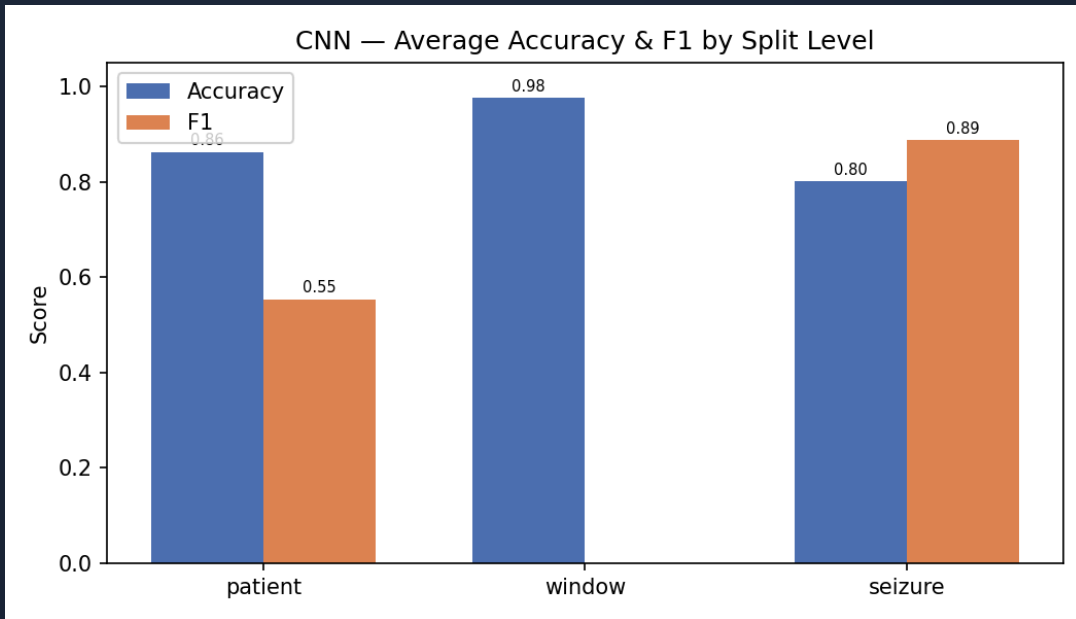
# Results: Window-Level CV Degeneracy

---

CNN and LSTM under window-level CV: **97-100% accuracy but 0% F1**

- Round-robin assignment creates unreliable per-fold class ratios
- Models learn to predict "non-seizure" for everything
- **Window-level CV is unsuitable** for this dataset

# Results: CNN Split-Level Comparison

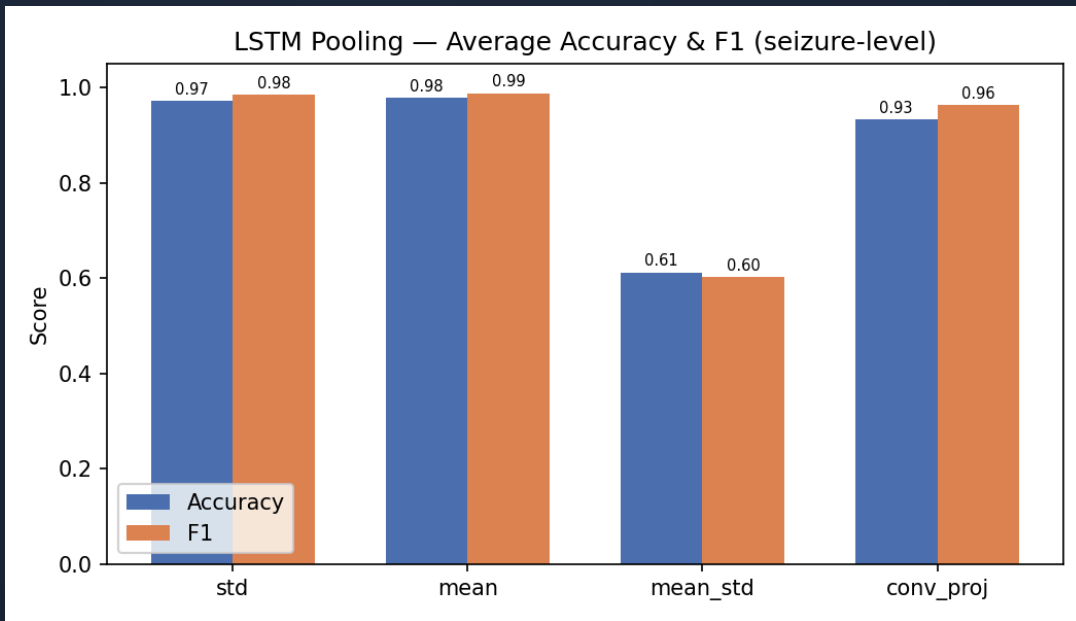


| CV Level | Avg Acc | Avg F1 |
|----------|---------|--------|
| patient  | 86.3%   | 0.554  |
| window   | 97.7%   | 0.000  |
| seizure  | 80.1%   | 0.888  |

“ Patient-level: meaningful but variable.  
Window-level: degenerate.  
Seizure-level: high precision, lower recall.

”

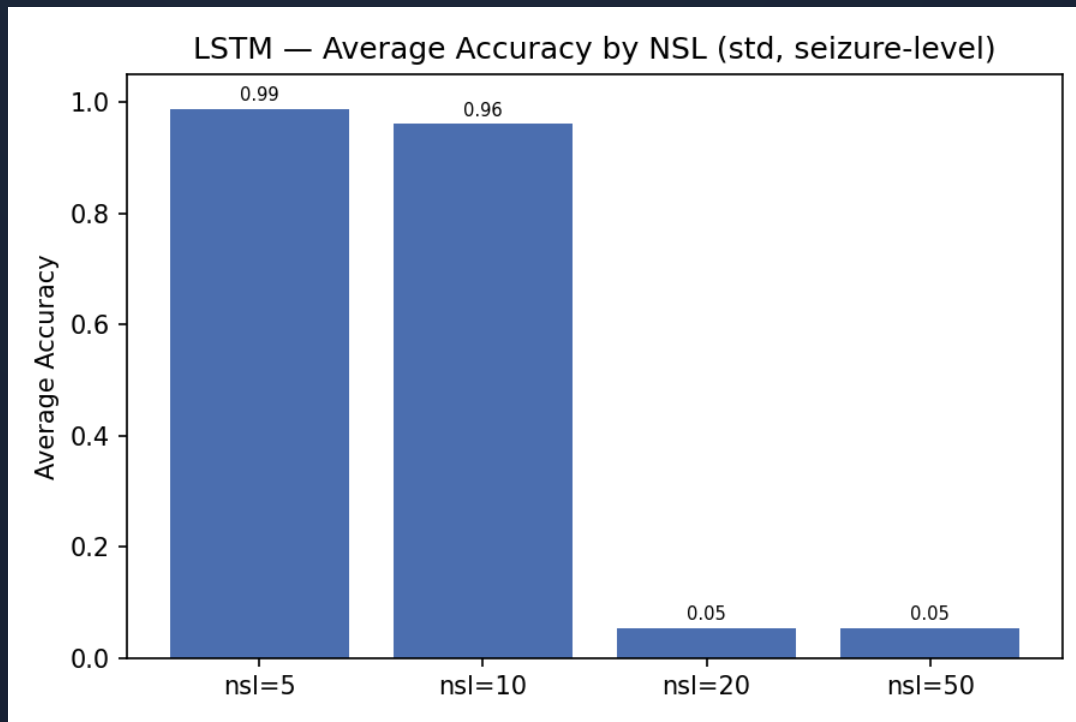
# Results: LSTM Pooling (Seizure-Level)



| Pooling          | Avg Acc      | Avg F1       |
|------------------|--------------|--------------|
| std (21-d)       | 97.2%        | 0.985        |
| mean (21-d)      | <b>97.8%</b> | <b>0.988</b> |
| conv_proj (32-d) | 93.3%        | 0.963        |
| mean_std (42-d)  | 61.2%        | 0.601        |

“ mean\_std unstable (2/5 folds fail) ”

# Results: Non-Seizure Length Sweep



| NSL | Avg Acc | Avg F1 |
|-----|---------|--------|
| 5   | 98.8%   | 0.994  |
| 10  | 96.1%   | 0.979  |
| 20  | 5.5%    | 0.000  |
| 50  | 5.5%    | 0.000  |

“

NSL  $\geq$  20  $\rightarrow$  training collapse

”



# Key Observations

---

- **LSTM (seizure-level)** dominates: 97.2% acc, 0.985 F1
- **CNN (patient-level)**: 86.3% acc but F1 only 0.554 — low recall on some patients
- **Window-level CV** produces degenerate results for all models
- **Threshold (75.5%)** outperforms RF (69.0%)
- **Mean & std pooling** both work well; mean\_std is unstable
- **NSL  $\geq 20$**  causes LSTM training collapse

# Conclusion

---

- LSTM (seizure-level) achieves best results: **97.2% acc, 0.985 F1**
- CNN (patient-level): 86.3% acc, 0.554 F1 — high variance across patients
- Window-level CV is degenerate — poor class ratios per fold
- Mean pooling slightly outperforms std for LSTM-std
- Should try LSTM-conv with more training
- **NSL  $\leq 10$**  is essential; NSL  $\geq 20$  causes collapse
- **Future:** remove pre/post-ictal windows, explore multichannel spatial info, longer temporal context

# **Thank You**

---

## **Any Questions?**